Azure Virtual Desktop Remote Access (Windows & macOS Clients) — Government (GCC High)

Implementation Solution Guide

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Executive Summary — Azure Virtual Desktop Remote Access (Government / Azure Government · GCC High)

The problem

Azure Virtual Desktop (AVD) on a **GCC High / Azure Government** tenant is **not** the Commercial portal with a different sign-in. The admin portals, the client feed URLs, the sign-in endpoint, and the session-host service endpoints are all **different** in the US sovereign cloud — and some capabilities **lag, are limited, or are absent** versus Commercial. Teams that follow a Commercial runbook verbatim typically end up with clients subscribed to the wrong feed and session hosts that fail to register because a `.us` endpoint is blocked. The failures are quiet, and they surface late.

The solution

This is the **Government (Azure Government / GCC High) edition** of the AVD Remote Access solution: AVD authored end to end for the US sovereign cloud. It uses the correct portals (`portal.azure.us`), the correct client feed (`rdweb.wvd.azure.us`), the correct Government service endpoints (`*.wvd.azure.us`, `*.usgovcloudapi.net`), and the identity model a Gov tenant actually supports — with **Commercial-vs-Azure-Government parity gaps called out** so nothing silently fails.

What's inside

- **Solution Overview** with an explicit *Commercial vs Azure Government — what's different* table.
- **Requirements Guide** — ten verifiable prerequisites (tenant, Government licensing, roles, resource provider, identity model, image, networking + the Gov endpoint allow-list, clients, pilot group, change control).
- **Implementation Guide** — ordered, click-to-implement steps in the Gov portals, following **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** with a gate at every stage and GCC-High caveats inline.
- **Testing & Troubleshooting** — twelve pass/fail test cases and a symptom→fix guide for the common sovereign-cloud failures.
- **Five PowerShell scripts** — Discovery, Validation, Report, Remediation (`-WhatIf`), and Set-ScopeTargeting — all on **US Gov** Azure/Graph endpoints.
- **GCC High parity reference** — cited, so you verify feature availability before you depend on it.

The outcome

AVD deployed **correctly on GCC High** — right portals, right endpoints, parity gaps handled — proven on a small pilot before you scale, expanded in controlled waves, with a documented rollback. Self-guided and tested, at a fraction of the cost of a bespoke consulting engagement.

Who it's for

Government, defense, and regulated organizations — and the MSPs and integrators serving them — deploying or correcting AVD on an Azure Government / GCC High tenant, who need the sovereign-cloud

specifics up front rather than learned the hard way.

Solution Overview — Azure Virtual Desktop Remote Access (Government / Azure Government · GCC High)

This is the **Government (Azure Government / GCC High) edition** of the Azure Virtual Desktop (AVD) Remote Access solution. It publishes Windows desktops and applications from session hosts running in **Azure Government** and delivers them to **Windows and macOS clients** — authored end to end for the US sovereign cloud, using the correct `*.us` admin portals, US Government service endpoints, and the identity model that a GCC High / Azure Government tenant actually supports.

The goal is a government tenant configured **correctly the first time** — not with Commercial steps that silently fail because the portal URL, the client feed URL, or a feature simply is not the same in the sovereign cloud.

> ***Naming convention (used throughout):*** *every object you create is named* `<Purpose> – <Scope> – <Stage>` *and reused verbatim. Example:* `AVD Host Pool – Finance – Pilot`, `AVD Workspace – Finance – Prod`, `AVD Users – Finance – Pilot`.

What this solution delivers

- A **host pool**, **workspace**, and **application group** deployed in Azure Government, published to a named group of pilot users first, then expanded.
- **Session hosts** (Windows 11 / Windows 10 Enterprise multi-session, or single-session) joined per your chosen identity model (Microsoft Entra join or AD DS / Microsoft Entra Domain Services).
- **Client access from Windows App / Remote Desktop client on Windows and macOS**, subscribed to the **Azure Government feed URL** — not the Commercial one.
- **Five PowerShell scripts** — Discovery, Validation, Report, Remediation (default `-WhatIf`), and Set-ScopeTargeting — all connecting to **US Gov** Azure and Microsoft Graph endpoints (`-Environment AzureUSGovernment` / `USGov`).
- A **Commercial-vs-Azure Government parity reference** (`gcc-high-parity-notes.md`) so you know, before you start, which capabilities differ or lag in the sovereign cloud.

Commercial vs Azure Government — what's different

AVD is a single product, but the sovereign cloud is **not the Commercial portal with a different sign-in**. The control plane, the admin portals, the client feed URLs, and the session-host service endpoints are all distinct. If you follow a Commercial runbook verbatim on a GCC High tenant, the most common outcome is that clients subscribe to the wrong feed, or session hosts fail agent registration because a `.us` endpoint is blocked. The table below is the short version; the fully cited detail is in `gcc-high-parity-notes.md`.

Area	Commercial	Azure Government / GCC High
Azure portal	`portal.azure.com`	`portal.azure.us`

Area	Commercial	Azure Government / GCC High
Microsoft 365 / Entra admin	`admin.microsoft.com`, `entra.microsoft.com`	`portal.office365.us`, `entra.microsoft.us`
Sign-in/ authentication endpoint	`login.microsoftonline.com`	`login.microsoftonline.us`
AVD client **feed URL** (subscribe)	`https://rdweb.wvd.microsoft.com` (or `.../api/arm/feeddiscovery`)	`https://rdweb.wvd.azure.us` (feed discovery: `https://rdweb.wvd.azure.us/api/arm/feeddiscovery?aadtenant=<yourdomain>`)
AVD **web client**	`https://client.wvd.microsoft.com/arm/webclient`	`https://rdweb.wvd.azure.us/arm/webclient/index.html`
Session-host service traffic FQDN	`*.wvd.microsoft.com`	`*.wvd.azure.us`
`ms-avd:` URI `env` value	`avdarm`	`usgov`
PowerShell / CLI environment	`AzureCloud` (default)	`AzureUSGovernment` (`Connect-AzAccount -Environment AzureUSGovernment`; Graph `Connect-MgGraph -Environment USGov`)
Feature availability	New features land here first	**Parity lags** — features release in Commercial first and arrive in Gov later, in preview, or not at all. Verify each feature for Azure Government before you rely on it.

The single most important takeaway: point admins at `portal.azure.us` and point clients at `rdweb.wvd.azure.us`. Those two substitutions prevent the majority of "it works in the demo tenant but not in ours" failures.

Sources: AVD is available in sovereign clouds including Azure for US Government ([Microsoft Learn — Prerequisites](https://learn.microsoft.com/en-us/azure/virtual-desktop/prerequisites)); Gov feed/web-client and session-host FQDNs ([Required FQDNs and endpoints — Azure for US Government tab](https://learn.microsoft.com/en-us/azure/virtual-desktop/safe-url-list?tabs=azure-for-us-government)); `env=usgov` URI value ([URI

schemes](https://learn.microsoft.com/en-us/azure/virtual-desktop/uri-scheme)).

The deployment model — Assess → Configure → Pilot → Validate → Expand

Every step in the Implementation Guide fits this model, and each stage has an **exit gate** — you do not proceed until the gate passes.

1. **Assess** — Confirm you are on a GCC High / Azure Government tenant, the licensing entitles AVD, the `Microsoft.DesktopVirtualization` resource provider is registered, roles are in place, and the required `*.us` endpoints are reachable. *(Gate: Discovery script reports all green; requirements in `02-requirements.md` all verified.)*

2. **Configure** — Create the named host pool, workspace, and application group in `portal.azure.us`; deploy session hosts on the chosen identity model; assign the pilot user group. *(Gate: resources exist with the agreed names; session hosts show **Available**.)*

3. **Pilot** — Publish to `AVD Users – <Scope> – Pilot` only; those users subscribe to the **Gov feed URL** from Windows App / Remote Desktop client on Windows and macOS. *(Gate: every pilot user connects to a desktop and launches a published app.)*

4. **Validate** — Run the Validation and Report scripts; confirm connectivity, profile behavior, and performance against the pass/fail criteria in `05-testing.md`. *(Gate: all test cases pass; no open Sev-1/Sev-2 issues.)*

5. **Expand** — Use `AVDG-Set-ScopeTargeting.ps1` to grow the assignment group in controlled waves, re-validating after each wave. *(Gate: each wave validated before the next is added.)*

Who this is for

Government, defense, and regulated organizations (and their MSPs / integrators) standing up or correcting AVD on a **GCC High or Azure Government** tenant, who need the sovereign-cloud specifics called out rather than discovered the hard way. Novice-friendly, click-to-implement, with full Azure Government navigation at every step.

What you need before you start

See `02-requirements.md` for the full, verifiable prerequisite list (layered on the Azure Solutions global requirements in `global-requirements-azure-solutions.md`). At a glance: a GCC High / Azure Government tenant, Gov-appropriate licensing (e.g., Microsoft 365 G3/G5 or Windows Enterprise E3/E5 Gov entitlement for AVD), the correct Azure roles, `portal.azure.us` access, and the `*.us` / `*.usgovcloudapi.net` endpoints allowed through your filtering.

Requirements Guide — Azure Virtual Desktop Remote Access (Azure Government / GCC High)

These are the **solution-specific** requirements, layered on top of the platform baseline in `global-requirements-azure-solutions.md` (G1–G6). Every requirement below is structured **Description / Why / Verify / Implement**. Where a Government/GCC High specific applies, it is called out inline; unconfirmed specifics are flagged **"verify for Azure Government."**

> *Reuse the naming convention `<Purpose> – <Scope> – <Stage>` for every object you create while satisfying these requirements.*

R1. Confirmed GCC High / Azure Government tenant

• **Description:** The tenant is a US sovereign-cloud tenant (Microsoft 365 GCC High / DoD paired with Microsoft Entra ID in Azure Government), not a Commercial or GCC (moderate) tenant.

• **Why:** The admin portals, sign-in endpoint, AVD feed URL, and session-host service endpoints are all different in the sovereign cloud. Every step in this solution assumes `*.us` portals and endpoints; running it against a Commercial tenant points clients at the wrong feed and blocks agent registration.

• **Verify:** Sign in at `https://portal.azure.us` and `https://portal.office365.us`. Confirm the sign-in redirects through `login.microsoftonline.us` and your primary domain resolves (GCC High tenants commonly use `<name>.onmicrosoft.us`). In `entra.microsoft.us` → **Overview**, confirm the tenant.

• **Implement:** If you are on Commercial/GCC and require GCC High, provision through an authorized AOS-G reseller (per global requirement G1). This is a procurement action, not a config change — flag it early.

> ***GCC High note (cited):*** *GCC High/DoD are built on Azure Government and use Microsoft Entra ID Gov; the `.onmicrosoft.us` domain and `login.microsoftonline.us` endpoint are used only for GCC High and DoD. See `gcc-high-parity-notes.md`.*

R2. AVD-eligible licensing (Government SKUs)

• **Description:** Every user connecting to AVD holds a license that entitles Azure Virtual Desktop, using the **Government** variants of the eligible SKUs.

• **Why:** AVD access rights are granted by the Windows/Microsoft 365 license; without an entitling SKU the deployment is not licensed and Windows on the session hosts must instead use per-user access pricing.

• **Verify:** In `portal.office365.us` → **Billing** → **Licenses**, confirm each in-scope user has an AVD-entitling SKU — e.g., **Microsoft 365 G3/G5** (Government), **Office 365 G3/G5** with a qualifying Windows Enterprise entitlement, or **Windows Enterprise E3/E5** (Government). Windows Server session hosts require **RDS CALs with Software Assurance** or **RDS User Subscription Licenses**.

• **Implement:** Procure the correct Government SKUs and assign them to `AVD Users – <Scope> – Pilot`. For external/commercial-purpose users, enroll an Azure subscription for per-user access pricing.

> *The eligible-SKU list (M365 E3/E5/A3/A5/F3/Business Premium; Windows Enterprise E3/E5; Windows Education A3/A5; Windows VDA) is documented for AVD generally; the **Government tenant uses the G-series equivalents**. Confirm your exact SKU entitlement with your licensing channel — **verify for Azure Government** that a given SKU is offered in your Gov agreement. Source: [Prerequisites —*

Operating systems and licenses](https://learn.microsoft.com/en-us/azure/virtual-desktop/prerequisites).

R3. Azure roles for AVD deployment (least privilege)

• **Description:** The operating accounts hold the RBAC roles needed to create AVD objects and session hosts, and to register the resource provider.

• **Why:** Host pool / workspace / application-group creation and VM deployment require sufficient RBAC; registering `Microsoft.DesktopVirtualization` requires the `*/register/action` operation (included in Contributor or Owner).

• **Verify:** In `portal.azure.us` → **Subscriptions** → **Access control (IAM)**, confirm the deploying account has **Contributor** or **Owner** on the target subscription/resource group, plus **Desktop Virtualization Contributor** (or the granular Host Pool/Application Group/Workspace Contributor roles) for AVD object management. In `entra.microsoft.us` → **Roles & admins**, confirm any directory roles needed for group/assignment work.

• **Implement:** Assign the minimum roles; prefer PIM-eligible activation where available in your Gov tenant. Name any custom role assignment scope-group `AVD Admins – <Scope> – Prod`.

R4. `Microsoft.DesktopVirtualization` resource provider registered

• **Description:** The AVD resource provider is registered on the target subscription in Azure Government.

• **Why:** AVD objects cannot be created until the provider is registered; this is a per-subscription action.

• **Verify:** In `portal.azure.us` → **Subscriptions** → **(your subscription)** → **Resource providers**, search **Microsoft.DesktopVirtualization** and confirm status **Registered**. (Scripted: `Get-AzResourceProvider -ProviderNamespace Microsoft.DesktopVirtualization` after `Connect-AzAccount -Environment AzureUSGovernment`.)

• **Implement:** Select the provider → **Register**, or run `Register-AzResourceProvider -ProviderNamespace Microsoft.DesktopVirtualization`. `AVDG-Remediation.ps1` performs this (default `-WhatIf`). Source: [Prerequisites](https://learn.microsoft.com/en-us/azure/virtual-desktop/prerequisites).

R5. Supported identity model chosen and reachable

• **Description:** One supported identity scenario is selected and functional: **Microsoft Entra-only** (session hosts Microsoft Entra joined, cloud users), **Entra ID + AD DS** (hybrid), or **Entra ID + Microsoft Entra Domain Services**.

• **Why:** Microsoft Entra ID always authenticates AVD users; session hosts must join the same Entra tenant or an AD domain reachable from the session-host VNet. Profile and SSO options depend on the model chosen.

• **Verify:** Decide the model and confirm prerequisites: for hybrid, Microsoft Entra Connect is syncing (UPN or SID matches across AD DS and Entra); for Entra-only, users exist in Entra ID (no personal Microsoft accounts). Confirm the session-host VNet can reach domain controllers/DNS if using AD DS or Entra Domain Services.

• **Implement:** Stand up the missing piece (Entra Connect for hybrid; a domain-join account **without MFA** for the join). Record the choice in the runbook.

__GCC High note:__ the supported identity scenarios (Entra-only, Entra + AD DS, Entra + Entra Domain Services) are the same set as Commercial. __Intune management__ of session hosts is supported in Azure Government (GCC, GCC High, DoD) for Entra-joined / hybrid-joined hosts, but __not__ for hosts joined to Microsoft Entra Domain Services, and Windows Server hosts can't enroll in Intune. Some newer identity features (e.g., __external identities__) reached Azure Government later than Commercial and parts remain in preview — __verify for Azure Government__ before relying on them. Sources: [Prerequisites — Identity](https://learn.microsoft.com/en-us/azure/virtual-desktop/prerequisites), [Intune for AVD](https://learn.microsoft.com/en-us/intune/solutions/azure-virtual-desktop).

R6. Supported session-host OS image (from the Gov Marketplace)

• **Description:** Session hosts use a supported 64-bit OS: Windows 11/10 Enterprise multi-session, Windows 11/10 Enterprise, or Windows Server 2016–2025.

• **Why:** Only these editions are supported; N/KN/LTSC editions, 32-bit, Arm64, Ultra/Ephemeral OS disks, and VM Scale Sets are **not** supported for session hosts.

• **Verify:** In `portal.azure.us` → Marketplace, confirm the intended image is available in your Gov region. Confirm the region hosts the compute you need.

• **Implement:** Select a supported image (or a custom image in an Azure Compute Gallery). Name the host pool `AVD Host Pool – <Scope> – Pilot`.

R7. Networking, region, and the Government endpoint allow-list

• **Description:** A VNet/subnet exists in the same Azure Government region as the session hosts, with outbound access (TCP 443) to the **US Gov** AVD service endpoints; no inbound ports required.

• **Why:** Session hosts establish an outbound reverse connection to the AVD service; if the Gov service FQDNs are blocked, agent registration and connections fail. The endpoints are the `.us` / `.usgovcloudapi.net` set, **not** the Commercial `*.wvd.microsoft.com` set.

• **Verify:** Confirm the VNet region matches the session-host region and RTT to that region is < 150 ms from users. Confirm your firewall/proxy allows the **Azure for US Government** endpoint set (see the allow-list below). The `AVDG-Validation.ps1` script and Microsoft's *Agent URL Tool* both check reachability.

• **Implement:** Create the VNet/subnet; open outbound 443 to the Gov endpoints; prefer the `WindowsVirtualDesktop` service tag and `AzureCloud`/`AzureMonitor`/`AzureActiveDirectory` tags for Azure Government where supported. Do **not** open inbound 3389.

Required Azure Government session-host endpoints (allow outbound):

Address (US Gov)	Port	Purpose
`*.wvd.azure.us`	443	AVD service traffic (incl. TCP RDP)
`*.service.windows.usgovcloud.microsoft`	443	Service traffic
`*.windows.usgovcloud.microsoft`	443	Service traffic

Address (US Gov)	Port	Purpose
`*.windows.usgovcloud-static.microsoft`	443	Service traffic
`login.microsoftonline.us`	443	Authentication
`*.prod.warm.ingest.monitor.core.usgovcloudapi.net`	443	Agent / diagnostics
`gcs.monitoring.core.usgovcloudapi.net`	443	Agent traffic
`azkms.core.usgovcloudapi.net`	1688	Windows activation
`mrsglobalstugviffx.blob.core.usgovcloudapi.net`	443	Agent / SxS stack updates
`wvdportalstorageblob.blob.core.usgovcloudapi.net`	443	Azure portal support
`ctldl.windowsupdate.com`	80	Certificates
`ocsp.msocsp.com`	80	Certificates

Required Azure Government client endpoints (allow outbound on user devices): `login.microsoftonline.us`, `*.wvd.azure.us`, `*.servicebus.usgovcloudapi.net`, plus the shared `go.microsoft.com`, `aka.ms`, `learn.microsoft.com`, `privacy.microsoft.com`, `*.cdn.office.net`, `graph.microsoft.com`, `windows.cloud.microsoft`, `windows365.microsoft.com`, `ecs.office.com`.

Source: [Required FQDNs and endpoints — Azure for US Government tab](https://learn.microsoft.com/en-us/azure/virtual-desktop/safe-url-list?tabs=azure-for-us-government). Note the two Azure fabric IPs (`169.254.169.254`, `168.63.129.16`) must **not** be proxied or redirected.

R8. Supported clients on Windows and macOS, pointed at the Gov feed

- **Description:** Pilot users have **Windows App** or the **Remote Desktop client** installed on Windows and/or macOS, subscribed to the **Azure Government** feed.
- **Why:** These are the supported clients; the RADC and MSTSC clients are not supported. On the sovereign cloud they must subscribe to the **Gov** feed URL, not the Commercial one.
- **Verify:** On Windows, subscribe using feed URL `https://rdweb.wvd.azure.us` (or feed discovery `https://rdweb.wvd.azure.us/api/arm/feeddiscovery?aadtenant=<yourdomain>`). On macOS, install Windows App from the Mac App Store → **Add Work or School Account** and subscribe to the same Gov feed. Web client: `https://rdweb.wvd.azure.us/arm/webclient/index.html`.
- **Implement:** Deploy the client; document the Gov feed URL in user instructions (`HELP_CENTER` / onboarding). For scripted/URI subscription use `ms-rd:subscribe?url=https://rdweb.wvd.azure.us` and for direct connect set the URI `env=usgov`. Sources: [URI schemes](https://learn.microsoft.com/en-us/azure/virtual-desktop/uri-scheme), [Required FQDNs — US Gov](https://learn.microsoft.com/en-us/azure/virtual-desktop/safe-url-list?tabs=azure-for-us-government).

R9. Named pilot user group and assignment plan

• **Description:** A named security group `AVD Users – <Scope> – Pilot` contains the initial pilot cohort; an expansion plan defines later waves.

• **Why:** Publishing to a small named group first contains blast radius and makes validation meaningful; scoped expansion keeps each wave testable.

• **Verify:** In `entra.microsoft.us` → **Groups**, confirm the pilot group exists and contains only pilot users. Confirm it is the group assigned to the application group.

• **Implement:** Create the group and assign it to the AVD application group. Use `AVDG-Set-ScopeTargeting.ps1` (default `-WhatIf`) to add later waves in a controlled way.

R10. Change control, backup, and rollback (per G5/G6)

• **Description:** An approved change window and a documented rollback exist for the deployment and each expansion wave.

• **Why:** AVD publishing changes are user-facing; a tested back-out protects the pilot cohort and satisfies governance.

• **Verify:** Confirm an approved change record, a scheduled window, and a written rollback (e.g., remove group assignment, deallocate hosts) before executing.

• **Implement:** Capture pre-change state with `AVDG-Discovery.ps1` / `AVDG-Report.ps1`; document the back-out; rehearse where feasible.

Implementation Guide — Azure Virtual Desktop Remote Access (Azure Government / GCC High)

Ordered, click-to-implement steps using the **Azure Government** portals. GCC-High / Azure-Government caveats are called out **inline at the step where they matter**. Follow the deployment model **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, and do not pass a stage until its **gate** is green.

> ***Before you touch anything:*** *you are working in the US sovereign cloud. Admin URL is* `portal.azure.us` *(not* `portal.azure.com`*), and clients subscribe to* `rdweb.wvd.azure.us` *(not* `rdweb.wvd.microsoft.com`*). Keep the naming convention* `<Purpose> – <Scope> – <Stage>` *and reuse names verbatim.*

Stage 1 — Assess

Step 1.1 — Confirm you are in Azure Government

1. Browse to `https://portal.azure.us` and sign in. Your sign-in must redirect through `login.microsoftonline.us`.

2. Top-right → confirm the correct **directory/tenant** (GCC High domains are often `<name>.onmicrosoft.us`).

3. Open `https://portal.office365.us` → **Billing** → **Licenses** and confirm AVD-entitling **Government** SKUs are assigned to your pilot users (R2).

Caveat: *If either portal bounces you to* `portal.azure.com` / `admin.microsoft.com`*, you are on a Commercial/GCC tenant, not GCC High — stop and confirm the tenant (R1) before continuing.*

Step 1.2 — Register the resource provider

1. In `portal.azure.us` → **Subscriptions** → select your subscription → **Resource providers**.

2. Search `Microsoft.DesktopVirtualization`. If status is **NotRegistered**, select it → **Register**. Wait for **Registered**.

Scripted equivalent (Gov): `Connect-AzAccount -Environment AzureUSGovernment` *then* `Register-AzResourceProvider -ProviderNamespace Microsoft.DesktopVirtualization`*.*

Step 1.3 — Confirm roles and endpoints

1. **Subscriptions** → **Access control (IAM)** → confirm your account has **Contributor/Owner** plus **Desktop Virtualization Contributor** (R3).

2. Allow the **Azure for US Government** session-host and client endpoints through your firewall/proxy (R7 table). Key ones: `*.wvd.azure.us`, `*.service.windows.usgovcloud.microsoft`, `login.microsoftonline.us`, and the `*.usgovcloudapi.net` agent/monitor endpoints.

3. Run `AVDG-Discovery.ps1` (read-only) to snapshot the tenant/subscription and confirm the provider, roles, and reachable endpoints.

GATE 1 (Assess): *Discovery reports the provider Registered, roles present, and Gov endpoints reachable; all* `02-requirements.md` *items verified. Do not configure until green.*

Stage 2 — Configure

Step 2.1 — Create the host pool

1. `portal.azure.us` → search **Azure Virtual Desktop** → **Host pools** → **Create**.

2. **Basics:** Subscription, Resource group (e.g., `rg-avd-<scope>-prod`), **Host pool name** = `AVD Host Pool – <Scope> – Pilot`, **Location** = your Azure Government region (this sets where host-pool *metadata* is stored).

Caveat: *the* ***Location*** *field must be an Azure Government region. Session-host compute can be in any Gov region, but keep it close to users (RTT < 150 ms).*

1. **Host pool type: Pooled** (multi-session) or **Personal** per your design. Choose a load-balancing algorithm (Breadth-first for pilot is typical).

2. **Validation environment:** set **No** for production pilots unless you intend to receive early service updates.

Step 2.2 — Add session hosts

1. On the **Virtual Machines** tab of the create flow (or later via **Session hosts** → **+**), set **Add virtual machines = Yes**.

2. **Image:** pick a supported OS from the **Gov** Marketplace (Windows 11 Enterprise multi-session recommended for pooled). Avoid N/KN/LTSC, 32-bit, Arm64, Ultra/Ephemeral OS disks, and VM Scale Sets (R6).

3. **Name prefix:** e.g., `avdsh-<scope>` (becomes `avdsh-<scope>-0`, `-1`, …).

4. **Identity — choose your model (R5):**

• **Microsoft Entra ID** (cloud-native): select **Microsoft Entra ID**; optionally tick **Enroll VM with Intune** (supported in Azure Government for Entra-joined hosts).

• **Active Directory** (AD DS / Entra Domain Services): select **Active Directory**, provide the domain-join UPN/OU.

> ***Caveat:*** *the domain-join account* ***must not have MFA enabled****, or the join fails. Hosts joined to* ***Microsoft Entra Domain Services cannot be managed by Intune****;* ***Windows Server*** *hosts cannot enroll in Intune at all.*

1. **Networking:** select the VNet/subnet from R7 (same Gov region). Do not assign public IPs; do not open inbound 3389.

2. **Domain/credentials:** provide the local admin (and domain-join, if AD) credentials. Create.

3. Wait until **Session hosts** shows each host **Status = Available**.

> ***Caveat (endpoints):*** *if a host stalls at* ***Needs assistance*** */ fails agent registration, the usual cause on Gov is a blocked* `.us` *endpoint. Re-check* `*.wvd.azure.us` *and the* `*.usgovcloudapi.net` *agent endpoints (R7), then run* `AVDG-Validation.ps1`*.*

Step 2.3 — Create the workspace and application group

1. **Application groups:** an app group (**Desktop** type for full desktop) is created with the host pool. Rename/confirm it as `AVD App Group – <Scope> – Pilot`.

2. **Workspaces** → **Create** → `AVD Workspace – <Scope> – Prod` → in **Application groups**, **register** the app group above.

> *A published resource only appears to users once its app group is* ***registered to a workspace*** *— a common "nothing shows up in the client" cause.*

> ***GATE 2 (Configure):*** *host pool, workspace, and app group exist with the agreed names; all session hosts* ***Available****; app group registered to the workspace.*

Stage 3 — Pilot

Step 3.1 — Assign the pilot group

1. `entra.microsoft.us` → **Groups** → confirm `AVD Users – <Scope> – Pilot` (R9) contains only pilot users.

2. In the AVD **Application group** → **Assignments** → **+** → add the pilot group. Assign only the pilot group at this stage.

Step 3.2 — Subscribe clients to the Government feed

This is the step most likely to go wrong if you reuse Commercial instructions. *Use the* ***Gov*** *feed URL.*

• **Windows (Windows App or Remote Desktop client):**

1. Install **Windows App** (Microsoft Store) or the **Remote Desktop client**.

2. **Subscribe with URL** → enter `https://rdweb.wvd.azure.us` (or feed discovery `https://rdweb.wvd.azure.us/api/arm/feeddiscovery?aadtenant=<yourdomain>`).

3. Sign in with the Gov work account (`user@<name>.onmicrosoft.us` or your Gov domain).

• **macOS (Windows App):**

1. Install **Windows App** from the **Mac App Store**.

2. Open it → **+** → **Add Work or School Account** → sign in with the Gov account → subscribe to the same `https://rdweb.wvd.azure.us` feed.

• **Web client (any OS):** browse to `https://rdweb.wvd.azure.us/arm/webclient/index.html`.

• **Scripted / URI:** subscribe with `ms-rd:subscribe?url=https://rdweb.wvd.azure.us`; direct connect with `ms-avd:connect?...&env=usgov&version=0`.

Caveat: *Do* ***not*** *use* `rdweb.wvd.microsoft.com` *or* `client.wvd.microsoft.com` *— those are Commercial and will not return Gov resources. The* `ms-avd: env` *value for Azure Government is* `usgov` *(Commercial is* `avdarm`*).*

Step 3.3 — First connection

1. Each pilot user opens the subscribed workspace, launches the **desktop**, and launches at least one **published app**.

2. Confirm resources appear and the session connects over TCP 443 (RDP Shortpath/UDP is an optimization, not a requirement).

GATE 3 (Pilot): *every pilot user (Windows and macOS as applicable) subscribes to the Gov feed, connects to a desktop, and launches a published app.*

Stage 4 — Validate

1. Run `AVDG-Validation.ps1` (read-only, Gov endpoints) — confirms provider, host-pool/session-host health, workspace/app-group registration, assignment, and endpoint reachability.

2. Run `AVDG-Report.ps1` to produce the deployment report (inventory + pass/fail summary) for your change record.

3. Walk the test cases in `05-testing.md`; resolve any failures.

GATE 4 (Validate): *all* `05-testing.md` *cases pass; Report shows no Sev-1/Sev-2 open items.*

Stage 5 — Expand

1. Plan waves (e.g., +25 users each). For each wave, run `AVDG-Set-ScopeTargeting.ps1` (default `-WhatIf`) to preview adding users to `AVD Users – <Scope> – Pilot` (or a `– Prod` group), review the plan, then re-run with `-Confirm`/`-WhatIf:$false` to apply.

2. Scale session hosts if needed (add hosts to the pool with the same name prefix); confirm each new host reaches **Available**.

3. Re-run `AVDG-Validation.ps1` after each wave.

GATE 5 (Expand): *each wave validated before the next is added; host capacity keeps pace with assigned users.*

Rollback (per R10 / G6)

- Remove the group assignment from the application group (users lose access immediately, hosts untouched).
- Deallocate/delete session hosts if backing out entirely.
- Unregister the app group from the workspace to hide resources without deleting them.
- Keep the pre-change `AVDG-Discovery.ps1` / `AVDG-Report.ps1` output as the restore reference.

Testing & Validation — Azure Virtual Desktop Remote Access (Azure Government / GCC High)

Run these test cases at **Gate 4 (Validate)** and after every expansion wave. Each case has an explicit **pass/fail**. Capture evidence (screenshots / script output) for your change record. Scripts referenced here connect to **US Gov** endpoints (`-Environment AzureUSGovernment` / `USGov`).

Tenant reminder: admin at `portal.azure.us`*, clients subscribe to* `rdweb.wvd.azure.us`*.*

T1. Tenant & environment is Azure Government

- **Steps:** Sign in to `portal.azure.us`; confirm redirect through `login.microsoftonline.us`; run `AVDG-Discovery.ps1`.
- **Pass:** Portal is `.us`, sign-in is `login.microsoftonline.us`, Discovery reports environment **AzureUSGovernment**.
- **Fail:** Any Commercial URL (`portal.azure.com`, `login.microsoftonline.com`) appears — you are on the wrong cloud; stop.

T2. Resource provider registered

- **Steps:** `AVDG-Validation.ps1`, or portal **Subscriptions** → **Resource providers**.
- **Pass:** `Microsoft.DesktopVirtualization` = **Registered**.
- **Fail:** NotRegistered → run `AVDG-Remediation.ps1` (review `-WhatIf` plan first).

T3. AVD objects exist with correct names

- **Steps:** Confirm host pool `AVD Host Pool – <Scope> – Pilot`, workspace `AVD Workspace – <Scope> – Prod`, app group `AVD App Group – <Scope> – Pilot`.
- **Pass:** All three exist, correctly named, app group **registered** to the workspace.
- **Fail:** Missing object or unregistered app group → resources won't appear in clients.

T4. Session hosts healthy

- **Steps:** AVD → **Session hosts**; or `AVDG-Validation.ps1`.
- **Pass:** Every host **Status = Available**, agent/SxS versions current.
- **Fail: Needs assistance / Unavailable** → almost always a blocked `.us` endpoint (T8) or a domain-join issue (MFA on the join account).

T5. Assignment scoped to pilot group

- **Steps:** App group → **Assignments**.
- **Pass:** Only `AVD Users – <Scope> – Pilot` assigned.
- **Fail:** Broader/other group assigned → correct before proceeding.

T6. Windows client subscribes to the Gov feed

- **Steps:** On Windows, subscribe with `https://rdweb.wvd.azure.us`; sign in with the Gov account.
- **Pass:** Workspace and resources appear.
- **Fail:** "We couldn't find any resources" → confirm the **Gov** feed URL (not `rdweb.wvd.microsoft.com`), correct account, and T3/T5.

T7. macOS client subscribes to the Gov feed

- **Steps:** On macOS, Windows App → **Add Work or School Account** → subscribe to `https://rdweb.wvd.azure.us`.
- **Pass:** Resources appear and a desktop connects.
- **Fail:** No resources → same checks as T6; confirm Windows App (not the legacy Microsoft Remote Desktop macOS app pointed at Commercial).

T8. Government endpoints reachable from session hosts

- **Steps:** `AVDG-Validation.ps1` endpoint check, and/or Microsoft's *Agent URL Tool* on a host.
- **Pass:** `*.wvd.azure.us`, `*.service.windows.usgovcloud.microsoft`, `login.microsoftonline.us`, and the `*.usgovcloudapi.net` agent/monitor endpoints all reachable on 443.
- **Fail:** Any blocked → open it in the firewall/proxy (R7); re-test.

T9. Desktop connection and published app launch

- **Steps:** As a pilot user, connect to the desktop; launch a published app.
- **Pass:** Desktop session established over 443; app launches; input/display normal.
- **Fail:** Connection drops or app won't launch → check host health (T4), profile/identity (T10), endpoints (T8).

T10. Identity / profile behavior for the chosen model

- **Steps:** Sign in as a pilot user; confirm SSO/prompt behavior and FSLogix profile load (if configured) matches the chosen model (R5).
- **Pass:** Sign-in succeeds per design; profile loads (if FSLogix configured); no repeated auth loops.
- **Fail:** Auth loops or profile errors → verify hybrid sync (UPN/SID match), or Entra-only profile prerequisites. **Verify for Azure Government** that any SSO/passwordless/external-identity feature you depend on is available in Gov (see `gcc-high-parity-notes.md`).

T11. Report generated

- **Steps:** `AVDG-Report.ps1`.
- **Pass:** Report lists inventory + pass/fail and shows **no open Sev-1/Sev-2**.
- **Fail:** Open Sev-1/Sev-2 → remediate before Gate 4 sign-off.

T12. Rollback rehearsed

- **Steps:** In a controlled window, remove then re-add the pilot group assignment.
- **Pass:** Access is revoked and restored as expected; back-out is documented.
- **Fail:** Rollback doesn't behave as documented → fix the runbook before expanding.

Exit criteria (Gate 4)

All of T1–T12 **pass**, evidence captured, and the Report shows no open Sev-1/Sev-2. Only then proceed to **Stage 5 — Expand**.

Troubleshooting — Azure Virtual Desktop Remote Access (Azure Government / GCC High)

Symptom → likely cause → fix, focused on the failures that are **specific to the US sovereign cloud**. Most Gov-only AVD problems come down to three things: the **wrong portal/feed URL**, a **blocked `.us` endpoint**, or a **domain-join/MFA** issue. Work top-down.

> *Admin at* `portal.azure.us`*; clients subscribe to* `rdweb.wvd.azure.us`*; scripts on* `AzureUSGovernment` */ Graph* `USGov`*.*

A. "We couldn't find any resources" in the client

• **Cause 1 — wrong (Commercial) feed URL.** The client was subscribed to `rdweb.wvd.microsoft.com` / `client.wvd.microsoft.com`.

• **Fix:** Remove the workspace and re-subscribe to `https://rdweb.wvd.azure.us` (feed discovery `https://rdweb.wvd.azure.us/api/arm/feeddiscovery?aadtenant=<yourdomain>`).

• **Cause 2 — app group not registered to a workspace.**

• **Fix:** AVD → **Workspaces** → **(your workspace)** → **Application groups** → **+** register `AVD App Group – <Scope> – Pilot` (see `03-implementation.md` Step 2.3 / test T3).

• **Cause 3 — user not assigned.**

• **Fix:** App group → **Assignments** → **+** add `AVD Users – <Scope> – Pilot` (T5).

• **Cause 4 — wrong account/tenant.** User signed in with a Commercial or wrong-domain account.

• **Fix:** Sign in with the Gov work account (`user@<name>.onmicrosoft.us` or your Gov domain) via `login.microsoftonline.us`.

B. macOS client shows nothing / won't add the account

• **Cause — legacy app or Commercial endpoint.** Using the old Microsoft Remote Desktop macOS app or a Commercial feed.

• **Fix:** Install **Windows App** from the **Mac App Store** → **+** → **Add Work or School Account** → subscribe to `https://rdweb.wvd.azure.us` (T7). Confirm the client endpoints `*.wvd.azure.us` and `*.servicebus.usgovcloudapi.net` are allowed on the device network.

C. Admin portal redirects to Commercial

• **Cause — wrong tenant / wrong portal.** You opened `portal.azure.com` or you're actually on a Commercial/GCC tenant.

• **Fix:** Use `portal.azure.us` / `portal.office365.us`. If sign-in still lands on `login.microsoftonline.com`, the tenant is not GCC High — re-confirm R1 before proceeding.

D. Session host stuck at "Needs assistance" / agent won't register

• **Cause 1 — blocked Gov endpoints (most common).** `*.wvd.azure.us` or the `*.usgovcloudapi.net` agent/monitor endpoints are blocked by firewall/proxy.

• **Fix:** Allow the full **Azure for US Government** session-host set (R7 / parity notes §3): `*.wvd.azure.us`, `*.service.windows.usgovcloud.microsoft`, `login.microsoftonline.us`, `*.prod.warm.ingest.monitor.core.usgovcloudapi.net`, `gcs.monitoring.core.usgovcloudapi.net`, `mrsglobalstugviffx.blob.core.usgovcloudapi.net`, etc. Re-run `AVDG-Validation.ps1` and Microsoft's *Agent URL Tool* (T8).

• **Cause 2 — fabric IPs proxied.** `169.254.169.254` / `168.63.129.16` are being intercepted/redirected.

• **Fix:** Ensure these are never proxied or routed away (they must reach the Azure fabric directly).

• **Cause 3 — registration token expired.**

• **Fix:** Regenerate the host-pool registration key and re-add the host.

E. Domain join fails during host deployment

• **Cause 1 — MFA on the join account.** The domain-join account has MFA enabled.

• **Fix:** Use a domain-join account **without MFA** (scoped, monitored). Re-deploy the host.

• **Cause 2 — no line of sight to DC/DNS.** The session-host VNet can't reach the domain controller/DNS.

• **Fix:** Confirm VNet/DNS routing to the DC (R5/R7); fix the OU path if specified.

• **Cause 3 — Entra Domain Services + Intune expectation.** You expected Intune management but hosts are Entra-Domain-Services-joined.

• **Fix:** Entra-Domain-Services-joined hosts **can't** be Intune-managed; switch to Entra-join or hybrid, or manage via GPO (parity notes §6).

F. Sign-in loops / SSO prompts unexpectedly

• **Cause 1 — hybrid identity mismatch.** UPN or SID doesn't match across AD DS and Entra ID.

• **Fix:** Correct the mismatch and re-sync Entra Connect (R5).

• **Cause 2 — SSO/passwordless feature not GA in Gov.** You configured an SSO/passwordless path that isn't available in Azure Government yet.

• **Fix: Verify for Azure Government** (parity notes §7) that the specific SSO/WHfB/FIDO2/Entra-Kerberos path is GA in your Gov tenant; fall back to a supported method.

G. Scripts connect to the wrong cloud / no data

• **Cause — default (Commercial) environment.** `Connect-AzAccount` / `Connect-MgGraph` ran without the Gov environment.

• **Fix:** Use `Connect-AzAccount -Environment AzureUSGovernment` and `Connect-MgGraph -Environment USGov`. All five scripts default to Gov; don't override to Commercial.

H. FSLogix profile won't load

• **Cause — identity/storage prerequisites for the model.** For Entra-join + Azure Files, users must be hybrid identities and the storage path configured accordingly.

• **Fix:** Match FSLogix prerequisites to your identity model (R5 / best practices). **Verify for Azure Government** that the Azure Files Entra-Kerberos / cloud-only path you're using is GA in Gov before relying on it.

I. A feature "should exist" but isn't in the Gov portal

• **Cause — parity lag.** The feature has shipped to Commercial but not yet to Azure Government.

• **Fix:** Check `gcc-high-parity-notes.md` and Microsoft's current Gov docs; treat recently-announced features as **verify for Azure Government**. Design around GA-in-Gov capabilities only.

Escalation checklist

1. Confirm cloud: `portal.azure.us` + `login.microsoftonline.us` (T1).
2. Confirm endpoints reachable: `AVDG-Validation.ps1` (T8).
3. Confirm objects/assignment: `AVDG-Discovery.ps1` (T3/T5).
4. Capture `AVDG-Report.ps1` output and attach to the ticket.

Best Practices — Azure Virtual Desktop Remote Access (Azure Government / GCC High)

Operating guidance for running AVD well on the US sovereign cloud. These complement the ordered steps in `03-implementation.md` and the checks in `05-testing.md`.

> *Admin at* `portal.azure.us`*; clients subscribe to* `rdweb.wvd.azure.us`*; PowerShell/CLI use* `AzureUSGovernment` */ Graph* `USGov`*.*

Sovereign-cloud discipline

• **Bookmark the `.us` portals** and remove Commercial bookmarks from the admin workstation to avoid muscle-memory mistakes: `portal.azure.us`, `portal.office365.us`, `entra.microsoft.us`, sign-in `login.microsoftonline.us`.

• **Standardize the Gov feed URL** in every piece of user documentation and onboarding: `https://rdweb.wvd.azure.us`. Treat any appearance of `rdweb.wvd.microsoft.com` as a bug.

• **Assume parity lag.** Before designing around any AVD feature, confirm it is GA in Azure Government — features ship to Commercial first. Keep `gcc-high-parity-notes.md` next to your design and re-verify at build time. Flag anything unconfirmed as **"verify for Azure Government."**

Naming & reuse

• Name every object `<Purpose> – <Scope> – <Stage>` and reuse verbatim: `AVD Host Pool – Finance – Pilot`, `AVD Workspace – Finance – Prod`, `AVD App Group – Finance – Pilot`, `AVD Users – Finance – Pilot`. Consistent names make the scripts, the report, and the rollback unambiguous.

Identity & access

• Prefer the **simplest identity model that meets policy** — Entra-only removes domain controllers from the critical path; use hybrid (AD DS) only where line-of-business apps require it.

• The **domain-join account must not have MFA** (it breaks the automated join) — scope it tightly and monitor it.

• Use **least privilege** for admins (Desktop Virtualization Contributor over Owner where possible) and PIM-eligible activation if available in your Gov tenant.

• **Never open inbound 3389.** AVD uses an outbound reverse connection. For host troubleshooting use just-in-time VM access, not a standing RDP port; don't assign public IPs to hosts.

Networking & endpoints

• Allow the **US Gov endpoint set** (R7) via **service tags** (`WindowsVirtualDesktop`, `AzureActiveDirectory`, `AzureMonitor`, `AzureCloud`) rather than hand-maintained IPs.

• **Never proxy or redirect** the Azure fabric IPs `169.254.169.254` and `168.63.129.16` — doing so breaks provisioning and health.

• Keep session hosts in the Gov region **closest to users** (RTT < 150 ms); enable RDP Shortpath where the network supports UDP for a better experience.

• Use **Azure Firewall's AVD protection** / FQDN tags to lock down outbound traffic without breaking the service.

Profiles & images

• Use **FSLogix** for pooled multi-session profiles; store containers on Azure Files/NetApp Files, and match the identity prerequisites for your model (hybrid identities for Entra-join + Azure Files). **Verify for Azure Government** that the specific Azure Files Entra Kerberos/cloud-only path you intend is GA in Gov before committing.

• Build a **golden image** in an Azure Compute Gallery for repeatable, patched hosts; avoid unsupported editions (N/KN/LTSC), 32-bit, Arm64, Ultra/Ephemeral OS disks, and VM Scale Sets.

Management & operations

• Use **Intune** for session-host policy where supported in your Gov tenant (Entra-joined / hybrid-joined hosts). Remember: **Entra Domain Services-joined** and **Windows Server** hosts can't enroll in Intune.

• Turn on **AVD Insights / diagnostics** to the Gov Log Analytics endpoints (`*.usgovcloudapi.net`) from day one — it's your evidence for T4/T9 troubleshooting.

• **Pilot small, expand in waves** with `AVDG-Set-ScopeTargeting.ps1` (default `-WhatIf`), re-validating after each wave. Scale host capacity to keep pace with assigned users.

Change control & safety

• Every deployment and expansion wave runs inside an **approved change window** with a **written rollback** (R10). Capture pre-change state with Discovery/Report first.

• Keep **Remediation and Set-ScopeTargeting on `-WhatIf` by default**; require an explicit confirmation to make changes.

• Rehearse the rollback (remove/re-add assignment) before the first real expansion.

Security & compliance posture

• Keep everything **inside the Azure Government boundary** — GCC High/DoD identity, data residency, and service dependencies are designed to stay in Azure Government; do not reach back to Commercial services for AVD dependencies.

• Enforce **Conditional Access / MFA for users** (not the domain-join service account) using the Gov Entra endpoints.

• Never store tokens, keys, or credentials in scripts or docs; the provided scripts are read-only or `-WhatIf` by default and hold no secrets.

Deliverables Summary — Azure Virtual Desktop Remote Access (Azure Government / GCC High)

Everything included in this Government (Azure Government / GCC High) edition, and how the pieces fit together.

> *This edition is authored for the US sovereign cloud: admin at* `portal.azure.us`*, clients at* `rdweb.wvd.azure.us`*, scripts on* `AzureUSGovernment` */ Graph* `USGov`*.*

Documents (source/)

Document	What it gives you
`01-solution-overview.md`	What the solution delivers, the deployment model, and the **Commercial vs Azure Government — what's different** table (portals, feed URLs, endpoints, parity).
`02-requirements.md`	Ten solution-specific requirements (R1–R10), each **Description / Why / Verify / Implement**, layered on the Azure global requirements — including the Government endpoint allow-list.
`03-implementation.md`	Ordered click-to-implement steps in the Gov portals, **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, with GCC-High caveats inline and stage gates.
`05-testing.md`	Twelve pass/fail test cases (T1–T12) for Gate 4 and each expansion wave.
`06-troubleshooting.md`	Symptom → cause → fix for the common Gov-specific failures (wrong feed, blocked `.us` endpoints, join/MFA, parity).
`07-best-practices.md`	Operating guidance: sovereign-cloud discipline, identity, networking, profiles, management, change control, compliance.
`gcc-high-parity-notes.md`	Cited Commercial-vs-Azure-Government reference: portals, endpoints, and feature-parity gaps to verify.
`executive-summary.md`	One-page summary for decision-makers.

Document	What it gives you
`global-requirements-azure-solutions.md`	Shared Azure platform baseline (G1–G6). *(Provided; not part of this authoring pass.)*

Scripts (scripts/)

All scripts connect to **US Gov** endpoints (`Connect-AzAccount -Environment AzureUSGovernment`; `Connect-MgGraph -Environment USGov`). Discovery, Validation, and Report are **read-only**; Remediation and Set-ScopeTargeting default to `-WhatIf`.

Script	Purpose	Mode
`AVDG-Discovery.ps1`	Snapshot the Gov tenant/subscription: provider registration, roles, AVD objects, session hosts, assignments.	Read-only
`AVDG-Validation.ps1`	Validate the deployment against pass/fail criteria, including Gov endpoint reachability.	Read-only
`AVDG-Report.ps1`	Produce a deployment report (inventory + pass/fail) for the change record.	Read-only
`AVDG-Remediation.ps1`	Fix common gaps (e.g., register the resource provider).	`-WhatIf` default
`AVDG-Set-ScopeTargeting.ps1`	Build/grow the assignment group in controlled waves.	`-WhatIf` default

How to use the set

1. **Assess** with `AVDG-Discovery.ps1` and `02-requirements.md`.
2. **Configure** per `03-implementation.md` in `portal.azure.us`.
3. **Pilot** — subscribe Windows/macOS clients to `rdweb.wvd.azure.us`.
4. **Validate** with `AVDG-Validation.ps1` + `05-testing.md`, capture `AVDG-Report.ps1`.
5. **Expand** in waves with `AVDG-Set-ScopeTargeting.ps1`, re-validating each wave.

Consult `gcc-high-parity-notes.md` whenever you rely on a specific feature, and `06-troubleshooting.md` when a step doesn't behave. Keep every change inside an approved window with a documented rollback (R10).

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

AVDG-Discovery.ps1

Purpose: Snapshots the current state of AVD in Azure Government so you can Assess before you Configure. Reports the connected cloud/environment, Microsoft.DesktopVirtualization registration, AVD host pools / workspaces / application groups, session hosts, and application-group assignments. Read-only — makes no changes. Replace placeholders like 'your-subscription-id' with your values. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\AVDG-Discovery.ps1 Modules: Az.Accounts, Az.Resources, Az.DesktopVirtualization, Microsoft.Graph.Authentication Scopes/sign-in (US Gov): Connect-AzAccount -Environment AzureUSGovernment ; Connect-MgGraph -Environment USGov Optional: -SubscriptionId your-subscription-id -ResourceGroupName your-resource-group EXPECTED RESULTS A console + object summary. PASS = environment is AzureUSGovernment, provider Registered, and AVD objects/hosts listed. FAIL/WARN lines flag a Commercial context, unregistered provider, or missing objects. .NOTES White-label — the repeatable way. Read-only against Azure/Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\AVDG-Discovery.ps1 Modules: Az.Accounts, Az.Resources, Az.DesktopVirtualization, Microsoft.Graph.Authentication Scopes/sign-in (US Gov): Connect-AzAccount -Environment AzureUSGovernment ; Connect-MgGraph -Environment USGov Optional: -SubscriptionId your-subscription-id -ResourceGroupName your-resource-group EXPECTED RESULTS A console + object summary. PASS = environment is AzureUSGovernment, provider Registered, and AVD objects/hosts listed. FAIL/WARN lines flag a Commercial context, unregistered provider, or missing objects. .NOTES White-label — the repeatable way. Read-only against Azure/Graph US Gov endpoints. #>

AVDG-Remediation.ps1

Purpose: Fixes the common prerequisite gaps this solution detects in Azure Government: registers the Microsoft.DesktopVirtualization resource provider, and (optionally) registers an application group to a workspace so published resources appear in clients. SAFE BY DEFAULT — runs in -WhatIf mode and only previews changes; pass -WhatIf:$false (or -Confirm) to apply. Replace placeholders like 'your-workspace-name' and 'your-app-group-name' with your values. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\AVDG-Remediation.ps1 Apply: powershell -ExecutionPolicy Bypass -File .\AVDG-Remediation.ps1 -WhatIf:$false Register app group: .\AVDG-Remediation.ps1 -WorkspaceName your-workspace-name -AppGroupName your-app-group-name -ResourceGroupName your-resource-group -WhatIf:$false Modules: Az.Accounts, Az.Resources, Az.DesktopVirtualization Sign-in (US Gov): Connect-AzAccount -Environment AzureUSGovernment EXPECTED RESULTS In -WhatIf mode: "What if:" lines describing each change, no changes made. With -WhatIf:$false: the provider becomes Registered and/or the app group is registered to the workspace, with confirmation output. .NOTES White-label — the repeatable way. Read-only against Azure/Graph US Gov endpoints; write actions gated by -WhatIf/-Confirm. #>

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\AVDG-Remediation.ps1 Apply: powershell -ExecutionPolicy Bypass -File .\AVDG-Remediation.ps1 -WhatIf:$false Register app group: .\AVDG-Remediation.ps1 -WorkspaceName your-workspace-name -AppGroupName your-app-group-name -ResourceGroupName your-resource-group -WhatIf:$false Modules: Az.Accounts,

Az.Resources, Az.DesktopVirtualization Sign-in (US Gov): Connect-AzAccount -Environment AzureUSGovernment EXPECTED RESULTS In -WhatIf mode: "What if:" lines describing each change, no changes made. With -WhatIf:$false: the provider becomes Registered and/or the app group is registered to the workspace, with confirmation output. .NOTES White-label — the repeatable way. Read-only against Azure/Graph US Gov endpoints; write actions gated by -WhatIf/-Confirm. #>

AVDG-Report.ps1

Purpose: Produces a shareable report of the Azure Virtual Desktop deployment in Azure Government for the change record: environment, provider status, host pools, session hosts and their health, workspaces, and application groups. Writes an HTML file and a CSV inventory. Read-only — makes no changes. Replace placeholders like 'your-resource-group' and 'your-output-folder' with your values. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\AVDG-Report.ps1 -ResourceGroupName your-resource-group -OutputPath your-output-folder Modules: Az.Accounts, Az.Resources, Az.DesktopVirtualization Sign-in (US Gov): Connect-AzAccount -Environment AzureUSGovernment EXPECTED RESULTS Console summary plus AVDG-Report-<timestamp>.html and AVDG-Inventory-<timestamp>.csv in the output folder. The report flags any session host not Available and any workspace with no registered application group. .NOTES White-label — the repeatable way. Read-only against Azure/Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\AVDG-Report.ps1 -ResourceGroupName your-resource-group -OutputPath your-output-folder Modules: Az.Accounts, Az.Resources, Az.DesktopVirtualization Sign-in (US Gov): Connect-AzAccount -Environment AzureUSGovernment EXPECTED RESULTS Console summary plus AVDG-Report-<timestamp>.html and AVDG-Inventory-<timestamp>.csv in the output folder. The report flags any session host not Available and any workspace with no registered application group. .NOTES White-label — the repeatable way. Read-only against Azure/Graph US Gov endpoints. #>

AVDG-Set-ScopeTargeting.ps1

Purpose: Manages the security group that scopes AVD access (for example 'AVD Users - Finance - Pilot') so you can expand the pilot in controlled waves. Ensures the group exists and adds the supplied users, previewing every change first. SAFE BY DEFAULT — runs in -WhatIf mode; pass -WhatIf:$false (or -Confirm) to apply. Replace placeholders like 'AVD Users - Finance - Pilot' and 'user@your-gov-domain' with your values. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\AVDG-Set-ScopeTargeting.ps1 -GroupName 'AVD Users - Finance - Pilot' -UserUpns 'user1@your-gov-domain','user2@your-gov-domain' Apply: .\AVDG-Set-ScopeTargeting.ps1 -GroupName 'AVD Users - Finance - Pilot' -UserUpns 'user1@your-gov-domain' -WhatIf:$false Modules: Microsoft.Graph.Authentication, Microsoft.Graph.Groups, Microsoft.Graph.Users Sign-in (US Gov Graph): Connect-MgGraph -Environment USGov -Scopes 'Group.ReadWrite.All','User.Read.All' EXPECTED RESULTS In -WhatIf mode: a plan of the group to create/use and each user to add, no changes made. With -WhatIf:$false: the group exists and the listed users are members, with confirmation output. Re-validate after each wave. .NOTES White-label — the repeatable way. Uses Microsoft Graph US Gov endpoints; write actions gated by -WhatIf/-Confirm. #>

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\AVDG-Set-ScopeTargeting.ps1 -GroupName 'AVD Users - Finance - Pilot' -UserUpns 'user1@your-gov-domain','user2@your-gov-domain' Apply: .\AVDG-Set-ScopeTargeting.ps1 -GroupName 'AVD Users - Finance - Pilot' -UserUpns 'user1@your-gov-domain' -WhatIf:$false Modules: Microsoft.Graph.Authentication, Microsoft.Graph.Groups, Microsoft.Graph.Users Sign-in (US Gov Graph): Connect-MgGraph -Environment

USGov -Scopes 'Group.ReadWrite.All','User.Read.All' EXPECTED RESULTS In -WhatIf mode: a plan of the group to create/use and each user to add, no changes made. With -WhatIf:$false: the group exists and the listed users are members, with confirmation output. Re-validate after each wave. .NOTES White-label — the repeatable way. Uses Microsoft Graph US Gov endpoints; write actions gated by -WhatIf/-Confirm. #>

AVDG-Validation.ps1

Purpose: Validates an Azure Virtual Desktop deployment in Azure Government against the solution's gate criteria: correct cloud, provider registered, host pool present with Available session hosts, app group registered to a workspace and assigned, and the required US Gov service endpoints reachable on 443. Read-only — makes no changes. Replace placeholders like 'your-host-pool-name' with your values. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\AVDG-Validation.ps1 -HostPoolName your-host-pool-name -ResourceGroupName your-resource-group Modules: Az.Accounts, Az.Resources, Az.DesktopVirtualization Sign-in (US Gov): Connect-AzAccount -Environment AzureUSGovernment EXPECTED RESULTS A checklist with [PASS]/[FAIL]/[WARN] per test and an overall result. Overall PASS = all gate checks pass; any [FAIL] (e.g., blocked *.wvd.azure.us, unavailable host) must be fixed before proceeding. .NOTES White-label — the repeatable way. Read-only against Azure/Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\AVDG-Validation.ps1 -HostPoolName your-host-pool-name -ResourceGroupName your-resource-group Modules: Az.Accounts, Az.Resources, Az.DesktopVirtualization Sign-in (US Gov): Connect-AzAccount -Environment AzureUSGovernment EXPECTED RESULTS A checklist with [PASS]/[FAIL]/[WARN] per test and an overall result. Overall PASS = all gate checks pass; any [FAIL] (e.g., blocked *.wvd.azure.us, unavailable host) must be fixed before proceeding. .NOTES White-label — the repeatable way. Read-only against Azure/Graph US Gov endpoints. #>